

Technical Supplement: Activation-Aligned Asynchronous Residual MPC

for Position-Controlled Manipulators with Learned Dynamics

Public evidence bundle for ROBIO 2026

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1 Purpose and evidence boundary

This supplement provides implementation, configuration, and diagnostic detail compressed in the conference manuscript. It introduces no results beyond the frozen or deterministically reanalysed public evidence bundle.

All results use MuJoCo position-controlled robot models. ABB IRB 2400 is the primary robot; UR5e is an independently trained and calibrated nominal within-robot replication. The evidence does not establish physical-robot performance, ABB hardware-torque feasibility, hard-real-time scheduling, robust-MPC guarantees, shared-model transfer, or arbitrary-morphology generalization. The post-freeze effort sweep is diagnostic only.

2 Controller and execution configuration

The controller is a one-layer GRU (hidden size 256; history length 16) that predicts velocity increments from aligned $[q_t, \dot{q}_t, q_t^{\text{ref}}]$ tokens, using one-step increment plus 20-step rollout loss on a 90/10 episode-disjoint train/validation split; recurrent windows remain within their assigned episode groups. Frozen controller cost weights/scales are shared; XML models, data distributions, checkpoints, normalizers, references, Robot-Spec contracts, and delays are robot-specific. The datasets are independently collected offline closed-loop position-actuator transitions driven by non-MPC reference excitations: hold, step, smooth-random, sinusoidal, delta-reference-random, and temporally correlated random references. The latter uses an autoregressive random-reference generator; no training rollout is generated by CEM or MPC. Workspace bounds, normalized excitation scales, and optimizer settings remain robot-specific. For UR5e, the frozen dataset contains 120,000 episodes of 1,000 control steps (120 million transitions) collected at normalized excitation scales 0.3, 0.5, 0.8, and 0.9. The complete machine-readable contracts are released with the evidence bundle.

Table 1: Frozen ABB controller settings. Per-joint vectors and all robot-specific artifacts are in the manifests.

Setting	Value	Setting	Value
Control interval	10 ms	History length	16
Horizon / samples / iterations	20 / 128 / 2	Elite ratio	0.08
Replan interval / delay D	5 / 6 steps	Guard	5 ms
Temporal discount	0.95	Projection	two-stage compiled
$w_q/w_{\dot{q}}/w_r/w_s$	1 / .10 / .20 / .05	$w_{\ddot{r}}/w_{\ddot{r}}/w_{\text{first}}$	.05 / .02 / .20
Joint / velocity barrier	10 / 5	Barrier max multiplier	2
Task position/orientation	1 / .25	Task scales	.05 m / .0873 rad
Feedback $K_q/K_{\dot{q}}/\delta r_{\text{max}}$	.3 / .015 / .015 rad	Residual maximum	(0.12,.10,.12,.15,.15,.20) rad

True-history window validation. Held-out model errors use fixed windows beginning at steps 16, 36, $\dots$, 176 of each saved 200-step rollout. Every window therefore has a complete 16-token ground-truth recurrent context before its first predicted successor. Each horizon is evaluated as a separate segmentwise RMSE: $h = 1$ measures the first successor and $h = 20$ pools prediction steps 1:20 within each window. These are not terminal errors. The historical start-of-rollout validation, which padded the missing initial GRU history, is retained in the evidence bundle for audit but is not used in the manuscript table.

Residual packets encode corrections, not reusable absolute commands. At packet age $\kappa = t - t_a$, the full protocol reconstructs and projects

$$q_t^{\text{cmd}} = \Pi_{v,\text{acc},\mathcal{Q}}(\bar{q}_t^{\text{IK}} + r_\kappa + \delta r_t; q_{t-1}^{\text{cmd}}, \dot{q}_{t-1}^{\text{cmd}}).$$

The launch-anchor replay counterfactual instead projects the cached planned absolute command q_κ^{plan} . Its large failure is therefore evidence for execution-time nominal-frame reanchoring required by an independently evolving nominal, not a competitive controller comparison.

Optional post-selection model-disagreement gate. The released controller exposes an optional gate that is disabled (`uncertainty_mode=off`) in every reported configuration. The planning GRU alone generates and scores all CEM candidates. Only after selection, independently trained, architecture-matched GRU dynamics models roll out the same selected H_u -step joint-reference sequence; all use the same non-MPC reference-excitation training protocol. If $\hat{x}_{h,d}^{(j)}$ is the prediction of model j at step h and state coordinate d , and s_d is the planning model’s training-state standard deviation, the gate records

$$u = \sqrt{\frac{1}{H_u n_x} \sum_{h=1}^{H_u} \sum_{d=1}^{n_x} \left(\frac{\text{std}_j \hat{x}_{h,d}^{(j)}}{s_d} \right)^2}.$$

This post-selection calculation neither rescales CEM candidates nor changes their costs, ranks, or packet activation step. In monitoring mode it is telemetry only. When the optional gate is enabled, a hysteretic rule can attenuate residual authority after sustained disagreement. An incomplete time-bounded evaluation, or high disagreement together with a predicted joint-margin, residual-saturation, or tracking-error-growth condition, invokes a strict fallback: both residual and packet feedback are set to zero, yielding the same Projected Direct IK command. A limited drift state can instead use current-reference feedback rather than packet-predicted feedback. Thresholds, model count, short horizon, and evaluation deadline are configurable implementation parameters; this paper does not calibrate or evaluate them and claims no OOD-detection capability, closed-loop benefit, or safety guarantee.

3 Artifacts, references, and calibration

The manifests record robot contracts, command limits, model/data identities, references, and activation delays. The public calibration payloads in `abb/calibration` and `ur5e/calibration` use

$$D = \left\lceil \frac{L_{\text{P95}} + g}{\Delta t} \right\rceil,$$

where L_{P95} is snapshot-to-publication latency, $g = 5$ ms, and $\Delta t = 10$ ms. Independent 500-plan calibrations gave ABB $L_{\text{P95}} = 52.28$ ms and $D = 6$ control steps, and UR5e $L_{\text{P95}} = 56.94$ ms and $D = 7$ control steps. For UR5e, `ur5e/contracts` additionally exposes the robot and reference contracts.

4 Full diagnostics

Table 2: Held-out open-loop model errors from complete-history windows, pooled over prediction steps $1:h$. σ_u is normalized reference-excitation standard deviation. Errors are $q/\dot{q}$ RMSE in rad/rads⁻¹; Div. is the divergent-window rate at $\sigma_u = 0.8$.

Robot, h	q RMSE, $\sigma_u = .5$	$\dot{q}$ RMSE, $\sigma_u = .5$	q RMSE, $\sigma_u = .8$	$\dot{q}$ RMSE, $\sigma_u = .8$	Div., $\sigma_u = .8$
ABB, 1	.000670	.001990	.001069	.005570	0%
ABB, 20	.004852	.034472	.006876	.051398	0%
UR5e, 1	.000132	.000465	.000160	.000532	0%
UR5e, 20	.001334	.002745	.001436	.003023	0%

Model validation and CEM ranking. Tables 2–3 provide the principal held-out and replay diagnostics. Complete window, rollout, excitation, and counterfactual-replay files are released in the evidence bundle. Replay covers retained, projection-consistent online candidates only; its lower branch error is not directly comparable with the broader random-excitation validation.

Table 3: ABB retained-candidate replay diagnostics. Spearman is computed within snapshots with defined ranks; all intervals are hierarchical bootstrap 95% intervals.

Metric	Estimate	95% interval
Within-snapshot Spearman	0.976	[0.917, 1.000]
Pairwise concordance	0.988	[0.957, 1.000]
Relative realized regret	0.192%	[0, 0.481]%
20-step branch q RMSE	0.889 mrad	[0.705, 1.121] mrad

Supporting planner diagnostics. Task-space reranking reduced deployed ABB TCP RMSE by 3.05 mm (95% CI [0.96, 4.32] mm) relative to joint-only final-pool scoring, while increasing end-to-end p95 latency by 6.82 ms. Two-stage compiled projection met its pre-specified 5% non-inferiority criterion against full compiled projection. These are accuracy–latency diagnostics for supporting execution mechanisms, not independent core tracking claims.

Robustness and timing. ABB robustness applies L3/L6 single-factor payload, actuator-gain, TCP-force, and observation-noise perturbations. Per-condition/bootstrap results, timing, packet behavior, projection activity, and paired Projected/Preview-IK comparisons are released with the evidence bundle. Primary experiments had no planner failure or imposed joint, command-velocity, or command-acceleration violation; this does not imply a physical actuator limit.

Mechanism isolation. The frozen 20-case ABB matrix gives +2.82 mm for removing activation alignment (95% CI [1.27, 4.39] mm), +654.60 mm for launch-anchor absolute replay ([528.68, 778.68] mm), and −0.95 mm for removing fast feedback ([−2.25, 0.28] mm), all relative to FullVirtual. A focused recurrent-history isolation changed TCP RMSE by −0.67 mm with an interval crossing zero. Thus the evidence supports activation consistency but does not establish an independent nominal tracking gain from history propagation. Supporting projection/task-score and post-freeze effort diagnostics did not replace the frozen primary configuration.